

Bhumika Goyal - Aspiring Data Scientist

Contact Information

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LinkedIn | GitHub

Objective

Dedicated and detail-oriented aspiring Data Scientist with experience in machine learning, geospatial data analysis, and predictive modeling. Seeking to leverage my skills in Python, SQL, and deep learning to contribute to innovative data science projects and drive data-driven decision-making.

Skills

Programming Languages: Python, R, Java, C

Data Science Tools: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras

Data Visualization: Power BI, Tableau

Databases: SQL, MongoDB, PostGIS, Neo4J, MySQL, PySpark

GIS Tools: ArcGIS, QGIS, ERDAS, Google Earth Pro

Machine Learning: NLP, Deep Learning, Classification, Regression, Clustering

Other Tools: Excel, Word, PowerPoint, R Studio, GenAI, AI Prompt Engineering, Blockchain Technology

Experience

Skysight.AI (Noida) | GIS Analyst

March 2024 - August 2024

- Analyzed geospatial data for AI-driven applications, integrating GIS data into machine learning pipelines.
- Assisted in the development of GIS-based AI models using Python, TensorFlow, and Scikit-learn.

- Collaborated with data science teams to enhance model performance and improve predictive accuracy.

Mentorless | Machine Learning Intern

February 2024 - March 2024

- Conducted data exploration and pre-processing using Python, Pandas, and NumPy.
- Developed and trained machine learning models using Scikit-learn and evaluated their performance.
- Gained hands-on experience with various machine learning algorithms, including Decision Trees, Random Forests, and K-Nearest Neighbors.

SAIRAD-IIGST | Research Intern

May 2022 - August 2022

- Conducted literature reviews and gathered data from diverse sources for research in data science and spatial analytics.
- Assisted in the design and implementation of data-driven research projects.
- Analyzed research data using Python, R, and relevant data analysis software.

Projects

Avalanche Prediction Model

Developed a deep learning model using TensorFlow and Keras to predict avalanches in the Pir Panjal Range, improving prediction accuracy by 15%.

Salary Estimation using K-Nearest Neighbor

Built a salary estimation model based on demographic and professional features using the K-Nearest Neighbor algorithm.

Household Electricity Consumption Prediction

Created a time series model using LSTM in TensorFlow to forecast household electricity consumption, achieving a 20% reduction in forecasting error.

Digit Recognition using Random Forest

Implemented a Random Forest classifier for handwritten digit recognition, achieving a 98% accuracy rate.

Titanic Survival Prediction using Naive Bayes

Applied the Naive Bayes algorithm to predict passenger survival on the Titanic with 85% accuracy.

Leaf Species Detection using Decision Tree

Built a Decision Tree model to classify leaf species based on morphological features, achieving high classification accuracy.

Customer Churn Prediction

Developed a classification model to predict customer churn using logistic regression and decision trees, aiding in customer retention strategies.

Credit Risk Analysis

Created a predictive model to assess credit risk using machine learning techniques, enhancing risk assessment by 25%.

Sentiment Analysis for Customer Reviews

Developed an NLP model to analyze sentiment in customer reviews, providing actionable insights for improving customer satisfaction.

Fraud Detection in Financial Transactions

Built a real-time fraud detection system using machine learning, reducing fraud incidence by 30%.

Certifications

Application of Satellite Altimetry for Inland Waterbodies - IIRS

Global Navigation Satellite Systems and Application - IIT Roorkee (NPTEL)

Earth Observation-Based Mapping, Monitoring, and Modelling of Landslides - IIRS

RS & GIS Application - IIRS

HackerRank SQL (Basic) - HackerRank

Fundamental Cloud Concepts Badge for AWS - Pluralsight

Education

Symbiosis International University Pune | M.Sc. in Data Science and Spatial Analytics

July 2023 - June 2025

Amity University Noida | B.Sc. in Geoinformatics

July 2020 - June 2023